

The complete anatomy of the oncology platform

One deployed service, 24 role portals, and a single patient record that moves from registration through infusion. This is the full map — architecture, every role, the end-to-end journey, and exactly what changed in this session.

24

ROLE PORTALS

~52

CLINICAL ENTITIES

4,796

LINES, CORE ROUTER

5

ENHANCEMENTS THIS SESSION

23 / 24

TESTS PASSING

01 Architecture

One FastAPI process serves both a general hospital-management layer and the oncology-specific layer, plus every front-end page — no separate static host, no serverless split. Confirmed against `render.yaml`: this is the only thing actually deployed.

TWO PRODUCT LINES, ONE SERVICE

The **general HMS** (Admin, ward Doctor/OPD, IPD nursing, Pharmacy, Inventory, Billing, TPA) is the older layer. The **CCA Oncology OS** is a distinct clinical domain sitting alongside it — its own patient identity (`CCAPatient`, never conflated with the general `Patient` table), its own schema, its own 15 role portals. This field guide covers the CCA layer, where every enhancement this session landed.

BACKEND MODULES

main.py

entry point

App wiring, default-org bootstrap, the 24-persona login registry, static serving of every HTML page.

auth.py

RBAC

Password policy, JWT issue/refresh, every `is_cca_*` role predicate, logout logic.

routers/cca.py**core**

The oncology API proper — patients, orders, staging, MDT, Treatment Plan, Treatment Order, Day-Care/Infusion.

4,796 lines

routers/cca_coordination.py

Financial cases, cross-department care coordination.

957 lines

routers/cca_oncology_ext.py

Radiation & Surgical Oncology prescriptions, the regimen library, Treatment Plan phases.

668 lines

routers/cca_diagnostics.py

Lab, pathology and radiology result capture and review.

models_cca.py**schema**

~45 tables: patients, orders, staging, MDT cases/decisions, Treatment Plan/Care Plan, Day-Care/Infusion administration.

1,054 lines

models_cca_oncology_ext.py

7 tables: RadiationPrescription, RadiationFraction, SurgicalPlan, Regimen, RegimenDrugLine, TreatmentPlanPhase, OncologyRecordExtension.

191 lines

rbac_projection.py

Per-role field tiers (FULL / CLINICAL_CONTEXT / FINANCE / MINIMAL / NONE) — the same Treatment Plan JSON is trimmed differently per viewing role.

migrations.py**ledger**

Additive-only ADD COLUMN ledger — the only way schema changes reach the live Postgres database, since there's no Alembic.

FRONTEND

24 static, vanilla-JS pages under `frontend/` — no build step, no framework. Every page loads a shared `js/api.js` for auth/token-refresh and role-aware navigation, and most CCA pages share `js/cca-summary.js` for the cross-role patient-summary panel. One page, `cca_os.html`, is retired dark-theme scaffolding kept only as a file — no login route reaches it.

02 Roles & portals

Every login lands on a dedicated page. 9 general-hospital roles, 15 oncology-specific roles.

ROLE	PORTAL	WHAT THEY DO
GENERAL HOSPITAL		
Admin	/admin.html	Org & user administration, medicine/lab custom-data
Doctor	/opd.html	Non-oncology OPD scribe workflow
Head Nurse	/headnurse.html	Ward KPI dashboard, shift scheduler, reports
Nurse	/ipd.html	Inpatient bedside care, vitals, nursing notes
Nursing Station	/frontdesk.html	Ward-level admission desk
Pharmacist	/pharmacy.html	Formulary, FEFO batch dispensing, narcotics register
Inventory Manager	/inventory.html	Procurement, GRN receiving, stock transfers
Billing	/billing.html	POS billing, itemised invoices
TPA	/tpa.html	Insurance pre-authorization, corporate claims
CCA ONCOLOGY OS		
Front Desk	/frontdesk.html	Oncology intake, consent capture, check-in
Primary Investigation ● renamed	/nurse_navigator.html	Patient oncology journey, cycle clearances, toxicity logs
Medical Oncologist	/medical_oncologist.html	Staging (AJCC), NCCN readiness, chemo regimens
Surgical Oncologist	/surgical_oncologist.html	Resection notes, MDT panel input
Radiation Oncologist	/radiation_oncologist.html	Radiation plans, fractional treatment logs
Radiologist	/radiologist.html	Imaging review, lesion measurement, RECIST response
Radiology Coordinator	/radiology_coordinator.html	Scheduling, image intake, scan-review coordination

Pathologist	/pathologist.html	Histopathology, IHC, biomarker sign-off
Lab / Phlebotomy	/laboratory.html	Sample collection, CBC/LFT/RFT pre-chemo clearances
Pharmacist (CCA)	/oncology_pharmacist.html	Treatment-order verification, preparation, dispensing
Day-Care / Infusion Nurse	/infusion_nurse.html	Chemotherapy administration, chair monitoring
MDT Coordinator	/mdt_coordinator.html	Tumour board scheduling, case discussion
External MDT Specialist	/external_mdt_specialist.html	Remote second opinions, case-scoped access
Patient Liaison	/patient_liaison.html	Patient communication, counselling, portal access
Financial Counsellor	/patient_financial_services.html	Treatment estimates, government schemes, insurance

03 Patient journey, end to end

One record moves through eight stages. Diamond markers flag exactly where this session's five enhancements sit inside that flow.

01 Registration Consent, ID/Aadhaar OCR, check-in Front Desk	02 Primary Investigation Vitals, ECOG/Karnofsky, pain (0–10), handoff Primary Investigation	03 OPD Consultation AI-Scribe note, diagnosis, orders Medical / Surgical / Radiation Onc.	04 Diagnostics Lab, imaging, histopathology results Lab, Radiologist, Pathologist
05 MDT / Tumour Board Optional case review & recommendation MDT Coordinator + treating clinician	06 Treatment Plan Draft, sign, authorize Medical / Surgical / Radiation Onc.	07 Treatment Order Per-cycle executable order, pharmacy readiness Oncologist + Pharmacist	08 Day-Care / Infusion Administration, monitoring, discharge Infusion Nurse

01	Registration Front Desk Patient registration, consent capture, ID/Aadhaar OCR, appointment check-in. Creates the CCAPatient record everything downstream hangs off.
02	Primary Investigation • renamed this session Primary Investigation (formerly "Nurse Navigator") Height/weight/BSA, vitals, ECOG performance status, Karnofsky score, Pain Score on a 0–10 scale, fall-risk, handoff note. This is a distinct 0–10 scale used at intake — not the new 0–5 Day-Care scale in stage 08. ◆ Enhancement 1 — label renamed everywhere it's shown; role ID and routing untouched

03

OPD Consultation & ordering

Medical / Surgical / Radiation Oncologist

AI-Scribe turns a dictated consultation into a structured note; detected diagnoses, prescriptions and lab tests populate as checkboxes the doctor confirms. The **Raise Order** card sends orders for Lab, Radiology, Pathology and Molecular Diagnostic work.

- ◆ **Enhancement 3 — Pathology, Radiology and Molecular Diagnostic added to a dropdown that previously only offered "Lab"; category now auto-suggested from the item name, always doctor-editable**

POST /api/cca/orders

CCAOrder.order_type

04

Diagnostics

Lab / Phlebotomy · Radiologist + Radiology Coordinator · Pathologist

Samples collected, imaging read with lesion measurements and RECIST response, histopathology and biomarker sign-off (ER/PR/HER2, IHC, molecular panels). Results post back onto the shared patient record every other role reads from.

05

MDT / Tumour Board ● optional, doctor's choice

MDT Coordinator · treating oncologist · External MDT Specialist

A complete case package (diagnosis, staging, imaging, history) compiles in one click, the board discusses and drafts a recommendation, and the treating clinician records a disposition — Accept, Partially Accept, or Reject with a reason. MDT review has never been a mandatory gate in this system; the treating doctor has always decided whether a case needs it.

- ◆ **Enhancement 2 — this stage now has a real, trackable ON/OFF switch instead of being implicit**

POST /api/cca/mdt/cases

POST /mdt/cases/{id}/recommendation

POST /mdt/cases/{id}/approve

MDTDecision.status

06

Treatment Plan — the authorization moment

Medical / Surgical / Radiation Oncologist only

The clinician-owned treatment strategy: intent, modality, protocol, planned sessions. Starts in DRAFT, carries no clinical authority until signed. Only a clinician of the plan's own modality may sign it — the three oncologist roles are never interchangeable here.

Pass to MDT — off → doctor signs directly, exactly as before

Pass to MDT — on →

signing blocked until an **APPROVED** or **PARTIALLY_APPROVED** MDT decision is linked

♦ **Enhancement 2 — requires_mdt flag + enforcement gate + a new "link MDT decision" step**

TreatmentPlan.requires_mdt

POST /treatment-plans/{id}/link-mdt-decision

POST /treatment-plans/{id}/sign

07

Care Plan & Treatment Order

Same oncologist roles · Pharmacist

A Care Plan can only reference an already-ACTIVE (signed) Treatment Plan — never the reverse. Each cycle gets its own Treatment Order; Pharmacy verifies product/expiry and moves it through Verified → Preparing → Ready → Dispensed → Received before Day-Care ever sees it.

08

Day-Care / Infusion delivery

Oncology Day-Care / Infusion Nurse — with read-only tracking for the treating doctor

Pre-treatment safety check, vascular access assessment, then one card per medication transcribed off the signed order — Start / Pause / Resume / Stop / Complete, each an append-only timestamped event.

♦ **Enhancement 4 — each medication card now prints its real Started / Completed time, not just a status badge; the treating doctor gets a new read-only status panel showing the same list without opening the Infusion Nurse's screen**

♦ **Enhancement 5 — the Monitoring tab gained a Pain Scale (0–5) and a VIP — Visual Infusion Phlebitis — Score (0–5), both nurse-observed, both optional, both purely documentation**

InfusionMedicationAdministration

InfusionMonitoringObservation.pain_score

.vip_score

GET /treatment/{order_id}/medications

04 This session, in full

Five requested enhancements, exactly as implemented — model fields, endpoints, and files touched.

01

● implemented

Nurse Navigator → Primary Investigation

SCOPE Display label only, everywhere it's shown to a user — page titles, sidebar labels, task-owner strings, role dropdowns, the login portal chooser.

UNCHANGED Internal role identifier `CCANurseNavigator`, the file `nurse_navigator.html`, every backend RBAC check.

FILES `nurse_navigator.html` `admin.html` `index.html` `frontdesk.html`
`medical_oncologist.html` `pathologist.html`
`radiation_oncologist.html` `surgical_oncologist.html` `js/api.js`
`main.py`

Pass to MDT checkbox

NEW FIELD

`TreatmentPlan.requires_mdt` — Boolean, defaults `False`

ON

Sends the complete case package to MDT via the existing 1-click compiler, then creates the plan tagged `requires_mdt: true`. `sign_treatment_plan` now rejects (409) until `mdt_decision_id` points at a decision with status `APPROVED` or `PARTIALLY_APPROVED`.

OFF

No MDT case created. Doctor develops and signs the plan directly — the same single-signature mechanism that already existed, now just explicit and untouched by any new gate.

NEW ENDPOINT

`POST /treatment-plans/{id}/link-mdt-decision` — attaches an approved decision once the board has reported back; doesn't bump the plan's version, since it isn't a protocol change.

TRACKING

An "MDT Pathway" / "Direct Authorization" badge sits next to every plan's status badge, so the two populations are visibly distinct, not just implicit in whether a case happens to have an MDT case attached.

SCOPE

Medical, Surgical and Radiation Oncologist only — the three roles the backend already restricts plan-signing to.

FILES

`models_cca.py` `migrations.py` `routers/cca.py` `rbac_projection.py`
`medical_oncologist.html` `surgical_oncologist.html`
`radiation_oncologist.html`

Lab / Pathology / Molecular Diagnostic ordering

BEFORE	The live "Raise Order" card's Order Type dropdown offered exactly one option: Lab.
AFTER	Radiology, Pathology and Molecular Diagnostic added. A deterministic keyword match — not an LLM call, matching how this codebase already corrects lab-test names — scans the manual item field and every checked AI-detected test, and pre-selects a category. Always editable; picking a value by hand hides the "auto-suggested" tag.
COURSE-CORRECTION	First built into <code>cca_os.html</code> , which turned out to be retired scaffolding no login route reaches. Caught, reverted, and rebuilt on the real live "Raise Order" card duplicated across all three oncologist pages.
SCOPE	Medical, Surgical and Radiation Oncologist — the roles that actually raise diagnostic orders. Radiologist paused pending your confirmation (see §06).
FILES	<code>medical_oncologist.html</code> <code>surgical_oncologist.html</code> <code>radiation_oncologist.html</code>

04

● implemented

Medication administration visibility

BACKEND

Already complete from an earlier pass — `InfusionMedicationAdministration` tracks per-drug status, start/end time and who administered it, via `GET /treatment/{order_id}/medications`.

NURSE-SIDE GAP CLOSED

Medication cards showed a status badge but never the actual timestamp. Now print `Started: <time>` / `Completed: <time>` directly on the card.

DOCTOR-SIDE GAP CLOSED

New read-only "Medication Administration Status" panel on all three oncologist pages — same name/dose/status/start-time list the nurse sees, no separate login required to check it.

FILES

`infusion_nurse.html` `medical_oncologist.html`
`surgical_oncologist.html` `radiation_oncologist.html`

05

● implemented

Pain Scale + VIP Score

NEW FIELDS

`InfusionMonitoringObservation.pain_score` and `.vip_score` — both Integer, 0–5, optional, range-validated only.

PAIN SCALE

Nurse's own observation of the patient's pain during infusion, 0–5 — distinct from the 0–10 scale already captured at intake (stage 02).

VIP SCORE

The standard, published Visual Infusion Phlebitis score (Jackson, 1998) for grading a peripheral IV site — 0 (no symptoms) through 5 (advanced thrombophlebitis, pyrexia present). Each point shown with its clinical descriptor in the dropdown itself.

BOUNDARY HELD

Pure documentation — nothing computes a decision or an alert from either score, consistent with this codebase's standing rule against invented clinical-safety logic.

FILES

`models_cca.py` `migrations.py` `routers/cca.py` `infusion_nurse.html`

Ad hoc

● implemented

Login page cleanup

REMOVED

The "Demonstration authentication only" disclaimer, and the "Need traditional password login?" toggle that hid a fully working password-login form behind a click.

KEPT WORKING

The real `POST /api/auth/login` password form is now shown directly. Cleaned up the JS that would otherwise have thrown on the now-removed toggle button and silently broken the form's own submit handler.

FILES

`index.html`

05 Governance boundaries every change respects

Standing rules for this codebase, held throughout — not introduced for this session.

No invented clinical logic

Nothing computes a dose, a threshold, or a safety flag anywhere in this session's work. Pain Scale and VIP Score are recorded, never interpreted.

Multi-tenant isolation

Every new endpoint checks the patient belongs to the caller's organization before returning or mutating anything, matching every existing CCA endpoint.

Durable audit trail

Every state change — MDT decision linked, medication event, monitoring observation — publishes a `DomainEvent`, the same append-only log the rest of the platform already relies on.

Role-tiered visibility

New fields (`requires_mdt`, `pain_score`, `vip_score`) were added to the correct RBAC projection tier, not exposed to every role by default.

06 Verification & current status

Backend modified files byte-compile cleanly models_cca.py · migrations.py · routers/cca.py · rbac_projection.py · main.py	● pass
Treatment Plan lifecycle & infusion workspace test suites tests/integration/test_treatment_plan_lifecycle.py · test_infusion_nurse_workspace.py	● 22 / 23 pass
5 new tests added for the MDT gate and pain/VIP scoring requires_mdt_enforcement, link-mdt-decision, cross-patient rejection, score range validation	● all pass
1 pre-existing failure test_queue_lists_todays_session_and_arrival_updates — a date/timezone issue, confirmed failing on a clean checkout before any change here	● unrelated
Git state 17 files modified, nothing committed	● uncommitted
Radiologist — order dropdown, MDT/Treatment Plan authoring Discussed, then explicitly paused before any code was written — nothing to revert, nothing built	● paused

CCA Oncology OS Field Guide — generated from direct source verification during this session. Every endpoint, model field and file path above was read from the working tree, not inferred.